

2ND MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USAR)
SECOND Semester, April/May, 2026

Paper Code: BS112**Subject: Applied Mathematics-II****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1. (a)	State the following: (i) First Shift property (ii) Second Shift property	2.5	3
(b)	Give the formula to find the Fourier series coefficients a_0 , a_n & b_n .	2.5	3
(c)	Find the Laplace transform of: $f(t) = \int_0^t [\sin(at) / t] dt$	2.5	3
(d)	Using the method of separation of variables, solve: $\partial u / \partial x = 2(\partial u / \partial t) + u$, where $u(x, 0) = 6e^{-3x}$.	2.5	4
2. (a)	Obtain the half range sine series for the function $f(x) = x^2$ in the interval $0 < x < 3$.	5	3
(b)	Find the Inverse Laplace transform of: $(s + 4) / [s(s - 1)(s^2 + 4)]$.	5	3
3. (a)	Give the solution of wave equation by D' Alembert's Method.	5	4
(b)	Find the temperature in a bar of length 2 whose ends are kept at zero & lateral surface insulated if the initial temperature is: $\sin(\pi x / 2) + 3 \sin(5\pi x / 2)$.	5	4
4. (a)	Use convolution theorem to find the inverse of the following: $1 / [s^2(s^2 + 1)]$.	5	3
(b)	A string is stretched & fastened to two points l apart. Motion is started by displacing the string in the form $y = a \sin(\pi x / l)$ from which it is released at a time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by: $y(x, t) = a \sin(\pi x / l) \cos(\pi ct / l)$.	5	4

****** End of Question Paper ******